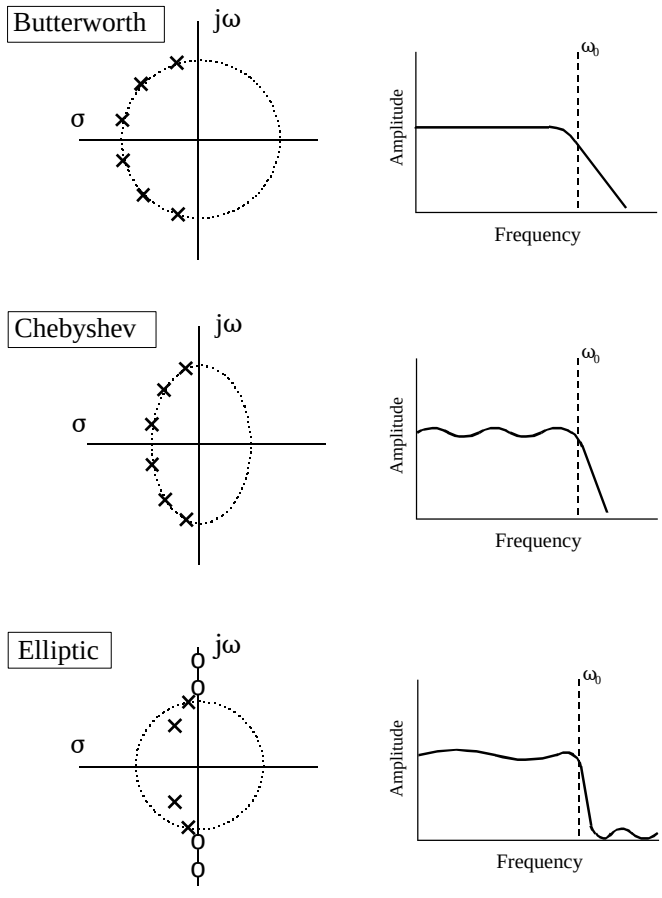


FIGURE 32-11
Classic pole-zero patterns. These are the three classic pole-zero patterns in filter design. Butterworth filters have poles equally spaced around a circle, resulting in a maximally flat response. Chebyshev filters have poles placed on an ellipse, providing a sharper transition, but at the cost of ripple in the passband. Elliptic filters add zeros to the stopband. This results in a faster transition, but with ripple in the passband *and* stopband.

Since each biquad produces two poles, **even order filters** (2 pole, 4 pole, 6 pole, etc.) can be constructed by cascading biquad stages. However, **odd order filters** (1 pole, 3 pole, 5 pole, etc.) require something that the biquad cannot provide: a single pole on the imaginary axis. This turns out to be nothing more than a simple RC circuit added to the cascade. For example, a 9 pole filter can be constructed from 5 stages: 4 Sallen-Key biquads, plus one stage consisting of a single capacitor and resistor.

These classic pole-zero patterns are for low-pass filters; however, they can be modified for other frequency responses. This is done by designing a low-pass filter, and then performing a mathematical transformation in the s-domain. We start by calculating the low-pass filter pole locations, and then writing the transfer function, $H(s)$, in the form of Eq. 32-3. The transfer function of the corresponding high-pass filter is found by replacing each "s" with "$1/s$", and then rearranging the expression to again be in the pole-zero form of Eq. 32-3. This defines new pole and zero locations that implement the high-pass filter. More complicated s-domain transforms can create band-pass and band-reject filters from an initial low-pass design. This type of mathematical manipulation in the s-domain is the central theme of filter design, and entire books are